

ChE 382: Unit Operations Laboratory

PROCESS CONTROL: LEVEL CONTROL

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Introduction

The Armfield PCT50 Level Control process unit provides a compact, hands-on introduction to the fundamental principles of process control engineering. The fluid-based process provides a clear demonstration of a single PID loop using Level as the measured variable. The effect of making changes to the system or to the controller configuration can be quickly investigated by applying repeatable disturbances to the process, clearly demonstrating the need for correct matching of system characteristics and controller settings. The software allows all appropriate measurements and controller settings to be displayed continuously and recorded using a data acquisition system for later analysis. For more information and theory on Proportional, Integral and Derivative controllers, refer to Experiment C in the Armfield software application for a brief explanation, as well as a textbook on Process Control.

PCT50 Level Control Process

Refer to the equipment diagram in the Armfield software application for more details about the PCT50 Level Process. The front view shows the layout of the PCT50 and the circuit diagrams show the process in operation with inflow to the process vessel (controlled by varying pump speed) or outflow from the process vessel (controlled by opening and closing a solenoid valve on a time cycle). We will be using the operation with inflow for our experimental procedure.

PCT50 is a level control process using water as the working fluid. Water stored in the lower sump tank is transferred to the upper process vessel via a submerged, variable speed centrifugal pump. A quick release connector allows the flexible pump outlet tube to be disconnected to aid priming of the pump after filling the sump tank with water. A vertical inlet arrangement in the process vessel allows visualization of the water entering the vessel, independent of water level and an integral non-return valve prevents water from draining back into the sump vessel when the pump speed is reduced or stopped. An inline ball valve above the quick release connector allows the flow of water entering the process vessel to be varied to suit a particular demonstration.

The water level inside the process vessel is measured using an electronic pressure sensor mounted through the floor of the vessel. One side of the pressure sensor is connected to the inside of the process vessel and the other is open to atmosphere, allowing the pressure in the process vessel to be measured relative to atmosphere. This sensor therefore measures the level of water inside the process vessel. The water level is given in millimeters (mm) in the mimic diagram. The water level can also be read using the analog scale (mm) on the side of process vessel.

Water drains from the process vessel back into the lower sump tank via two outlets on the floor of the process vessel. Flow through the second outlet valve can be started and stopped by a solenoid valve under remote control. Both outlets incorporate hand operated ball valves that allow the flow of water to be varied continuously to suit a particular demonstration. Both outlets also incorporate an interchangeable orifice that allows the flow to be fixed at predetermined values, varying only with the size of the orifice and the head in the process vessel. Alternative sizes of orifice are stored in threaded holes at the front of the base plate when not in use.

An overflow in the process vessel returns water to the sump tank, preventing the process vessel from flooding in use. The pressure sensor measuring the level, the centrifugal pump and the solenoid valve are connected to an electrical interface allowing the process to be operated directly from a PC.

The computer software supplied with PCT50 allows control of the level process and data logging of the responses.

Initial Filling of the Sump Tank and Priming the Pump

Ensure that the drain valve at the rear of the sump tank is fully closed before filling the sump tank. The sump tank can be filled by pouring deionized water through the cutout at the front or via the process vessel at the top. The deionized water level should be approximately 30 mm below the cut out in normal use. The unit is drained using the ball valve located at the rear.

Priming the Pump: To ensure reliable priming of the water pump after filling it for the first time or after draining and refilling the equipment, the flexible tube from the pump outlet can be temporarily disconnected from the system by pressing the latch on the quick release connector. After running the pump briefly at high speed, with the tube disconnected to expel any trapped air, the tube can be reconnected for normal operation.

EXPERIMENTS:

Three experiments will be performed:

Experiment 1- Manual control of water level (open loop)

Experiment 2- PID control of level, closed loop, **P control only**

Experiment 3- PID control of water level, closed loop, **P + I control**

Experiment 1: Manual control of water level, open loop

Note: It is recommended to perform Experiment 1 during the prelab period in order to have sufficient time to complete all the experiments.

The objective of this experiment is to determine; a) pump speed required to maintain a constant level in the process vessel, b) effect of inflow and outflow disturbances on the water level, and c) water flow rate when draining and filling the process vessel. This will be done by controlling the level of water in the process vessel by manually varying the pump speed that is supplying the water to the vessel and using a solenoid valve to vary the inflow and outflow of water to the vessel.

Method

Using the PC to operate the process, the pump speed (water flow into the process vessel) will be varied to reach and maintain a steady water level in the process vessel while water is continuously draining through the outflow valve. Disturbances will also be applied to the process by opening and closing a solenoid valve, resulting in a change to the water outflow and a water level change in the process vessel. The process vessel will also be drained to determine the water flow rate and filled at different pump speeds to determine the water flowrate. A pressure sensor is used to measure the water level in the process vessel during the experiments.

Experimental Setup

Turn on the computer (PC) and connect the power cable to the controller (black box to the left of the vessel). Start the PCT50 software and select Experiment 1: Inflow Control (PC control). Ensure that the Virtual COM port has been enabled using the red telephone icon. Ask the instructor for help if the COM port is not enabled.

Fill the sump tank with deionized water to the correct level (see above). Fully open the outlet valve CV2 and fit a 4mm diameter orifice below it. Fit a 2mm orifice below the solenoid valve. Prime the pump if needed following the procedure from above.

Initial Adjustment of Inlet Valve CV1:

It is advisable to adjust the inlet valve CV1 to give the appropriate water flow into the process vessel during the subsequent experiments:

- Fully open outlet valve CV2 and fully open inlet valve CV1.
- Choose **View/Diagram** in the software menu bar to display the mimic diagram of the process.
- Click **Power On (in Controls)**.
- Check the water level on the mimic diagram. If it is not zero, ask the instructor for assistance in zeroing the reading.
- Set the pump speed to 50% by typing in the value or by clicking the up arrow. Water will start flowing into the process vessel. If nothing happens, you may need to go into the PID controller box on the mimic diagram and make sure the Mode of Operation is on MANUAL and click on APPLY .
- As the water level rises, adjust the inlet valve CV1 until the level remains steady at approximately mid height in the process vessel.
- Reset the pump speed to 0% on the mimic diagram and allow the vessel to drain via the outlet valve CV2.

Experimental Procedure

Initial Level Control

1. Starting with an empty process vessel, gradually increase the pump speed and observe that the level rises when the water flow into the vessel exceeds the water flow out. Observe that the level remains relatively constant if the pump speed matches the outflow. Record the pump speed at that level. With the level relatively constant in the process vessel, press the “GO” icon to begin the data acquisition to record the water level vs time at that pump speed. Record data for about 5 minutes, then go to the next step, keeping the data acquisition running.

Disturbances

2. With the pump speed and level at the values from above, open the solenoid valve by clicking the button on the mimic diagram (1 indicates that the valve is open). Observe that the level falls slowly due to the additional outflow from the solenoid valve. When the level has fallen by at least 20mm, note the current pump speed, then increase the pump speed to restore and maintain the original water level for several minutes. Note the new pump speed.
3. Close the solenoid valve by clicking the button on the mimic diagram (0 indicates that the valve is closed). Observe that the level rises very slowly because the flow now exceeds the outflow. When the level has risen by at least 20mm, note the current pump speed and then reduce the pump speed to restore the water level to its original level. After several minutes, reduce the pump speed to 0% and allow the process vessel to drain. Keep recording the data using the data acquisition system until the process vessel is completely drained and then choose the “STOP” icon to end the data acquisition. Save your data from experiments 1-3 as an excel file.

Water Fill and Drain Rate

4. Create a new results sheet in the toolbar and then choose “GO” to start the data acquisition. Gradually increase the pump speed to a value greater than in step 1 (Initial Level Control) and the level starts to rise. Leave the pump at this speed and allow the vessel to fill. When the level reaches 100mm, stop the pump and allow the vessel to drain. When the vessel is empty, start the pump at 75% speed and allow the vessel to fill again. When the level reaches 100mm stop the pump and allow the vessel to drain. When the vessel is empty, stop the data acquisition and save your data as an excel file.

Results and Discussion:

Graph your results from the excel files:

Initial level control: Water level versus time at the 1st pump speed

Disturbances:

Water level versus time (solenoid on at 1st pump speed and 2nd pump speed)

Water level versus time (solenoid off at 2nd pump speed and 3rd pump speed).

Water flow rate-drain and fill:

Water level versus time during filling (note the pump speed)

Water level versus time during draining (0% pump speed)

Water level versus time during filling (75% pump speed)

Briefly describe what is happening in each experiment and why. Report the pump speeds needed to maintain the water level and the increase or decrease of the pump speed due to the different disturbances. What were the water filling flowrates at the two different pump speeds? Did the water draining flowrate change or stay the same? Is it easy or hard to maintain a constant water level with manual control?

Experiment 2: PID control of level, closed loop, P control only

The objective of this experiment is to determine the effect of different Proportional Band settings (P) on the response (offset or actual level versus setpoint level) and stability of a level process using a P (proportional band)- only controller that varies pump speed.

Method

The first set of experiments will start with the outflow valve (CV2) fully open. In this operation, water will drain continuously, with the flow rate determined by the size of the orifice fitting. Using a computer to operate the process, pump speed will be varied automatically using a proportional controller to maintain a steady water level in the process vessel. Trials will be performed to determine the offset (actual level – setpoint level) and stability at different P values.

A second set of experiments will introduce a disturbance to the process by opening and closing the solenoid valve, resulting in a change in the outflow and level in the process vessel. The setpoint will also be varied. Pump speed will again be varied automatically using the proportional controller to maintain a steady water level in the process vessel. Trials will be performed to determine the offset at different P values for the disturbances.

Experimental Setup

Follow the setup instructions given in Experiment 1 to turn on the unit, start the software, fill the sump tank with water, adjust valve CV1 and ensure that the pump and level (pressure) sensor are working correctly. Ask the instructor of TA for help if needed.

Experimental Procedure: P only controller (7 Proportional Band Settings)

Choose the PID box on the mimic diagram and set the Proportional Band (P) to 200%, the Integral Time I to 0, and the Derivative Time D to 0. Set the Set Point to 75mm (default at setup). Click APPLY to enter the changes to the settings.

1. Choose Automatic Mode of Operation and click APPLY. Go back to the mimic diagram and select the “GO” icon to begin the data acquisition. The pump will start, and the level will gradually increase, and water will drain from the vessel via the outlet valve CV2. The level will continue to rise until it reaches a somewhat steady level, however this will not correspond exactly to the setpoint of 75mm because there is an inherent offset when using a P-only controller. Wait a few minutes, record the actual water level and note the stability of the level.
2. Open the solenoid valve by clicking the button on the mimic diagram (1 indicates open valve) to disturb the process. Observe that the level falls slightly, pump speeds up slightly due to the increased error, but a large offset remains. When the level was settled to a relatively constant value, note the stability of the level and record the water level. Close the solenoid valve and allow the level to settle back to the value from step 1.
3. Adjust the Setpoint to 100mm and click APPLY. Observe the response to the requested increase in water level. When the level has settled to a relatively constant value, record the water level, note the stability, and then return the Setpoint to 75mm and observe the response.

The next 6 trials will repeat Steps 1-3 above for 6 different P values:

100%, 50%, 20%, 10%, 5%, 2%

At the end of all your trials, select the “STOP” icon to end the data acquisition. Save your data from your trials as an excel file.

NOTE: For high settings of the Proportional Band, such as 200%, the control action may be too conservative when responding to the error in the level. This will result in a very slow response, or possibly no response. In that case, there will be no water inflow into the process vessel. If this happens, wait a few minutes and then go to the next Proportional Band value.

Results and Discussion:

Graph your results from the excel files:

Water level versus time at the 7 Proportional Band values

Water level versus time for the increase outflow disturbance

Water level versus time for the change in Setpoint (100mm)

Calculate the offset for each Proportional Band value, disturbance, and Setpoint change and make a table to show the trend. Discuss the stability of the water level for each P value. Based on your results, choose a P value that gives stable control with a reasonably small offset. You will use this value in the next experiment with the P+I controller.

Experiment 3: PID control of level, closed loop, P + I control

The objective of this experiment is to determine the effect of different Integral Time settings (I) at a constant P setting on the response (offset or actual level versus setpoint level) and stability of a level process using a P + I controller that varies pump speed.

Method

The first set of experiments will start with the outflow valve (CV2) fully open. In this operation, water will drain continuously, with the flow rate determined by the size of the orifice fitting. Using a computer to operate the process, pump speed will be varied automatically using a P + I controller to maintain a steady water level in the process vessel. Trials will be performed to determine the offset (actual level – setpoint level) and stability at different I times.

A second set of experiments will introduce a disturbance to the process by opening and closing the solenoid valve, resulting in a change in the outflow and level in the process vessel. The setpoint will also be varied. Pump speed will again be varied automatically using the P + I controller to maintain a steady water level in the process vessel. Trials will be performed to determine the offset at different I values for the disturbances.

Experimental Setup

The apparatus should already be set up from Experiment 2. If not, follow the setup instructions given in Experiment 1 to turn on the unit, start the software, fill the sump tank with water, adjust valve CV1 and ensure that the pump and level (pressure) sensor are working correctly. Ask the instructor of TA for help if needed.

Experimental Procedure: P + I controller (Constant P, Multiple Integral Time Settings)

Create a new results sheet for this experiment in the software toolbar.

Choose the PID box on the mimic diagram and set the Proportional Band (P) to the value selected from Experiment 2, the Integral Time I to 0 seconds, and the Derivative Time D to 0. Set the Set Point to 75mm (default at setup). Click APPLY to enter the changes to the settings.

1. Choose Automatic Mode of Operation and click APPLY. Go back to the mimic diagram and select the “GO” icon to begin the data acquisition. Allow the water level to stabilize at a constant level. Adjust the I value in the PID box to **100 seconds** and click APPLY. Observe that the pump speed varies to reduce the offset. When the level has settled to a relatively constant value, record the water level and the stability of the level with time.
2. Open the solenoid valve to disturb the process. Observe that the level falls slightly, the pump speed varies, and the level gradually returns to a value near the set point. Wait a few minutes, record the actual water level and note the stability of the water level with time. Close the solenoid valve and allow the level to settle back to the value from step 1.
3. Adjust the Setpoint to 100mm and click APPLY. Observe the response to the requested increase in water level. When the level has settled to a relatively constant value, record the water level, note the stability, and then return the Setpoint to 75mm and observe the response.
4. The next set of trials will repeat Steps 1-3 above for different I values, starting with 50 seconds. Continue reducing the Integral time setting and repeating steps 1-3 until the level becomes unstable.

At the end of all your trials, select the “STOP” icon to end the data acquisition. Save your data from your trials as an excel file.

Results and Discussion:

Graph your results from the excel files:

Water level versus time at the different Integral Time and constant P values

Water level versus time for the increase outflow disturbance

Water level versus time for the change in Setpoint (100mm)

Calculate the offset for each Integral Time value, disturbance, and Setpoint change and make a table to show the trend. Discuss the stability of the water level for each I value. When does the water level control become unstable?

General Comments and Discussion Points

A large proportional band setting will result in very stable control, but will also result in slow changes and large offsets in the water level.

Reducing the proportional band will improve the speed of the response and give smaller offsets, but the process may become unstable if P is reduced too far.

Optimum results are obtained with the lowest P band settings that gives stable control but with the addition of Integral control, will eliminate the offset.

Reducing the integral time setting increases the speed at which the offset is reduced, but if I is too short, it will result in instability because the process is not able to respond quickly enough to the changes from the controller. This effect is called integral saturation, or integral wind up, and should be avoided.

Increasing the Proportional Band reduces the effect of Integral control, so a shorter I time is required to eliminate the offset more quickly.